



improves scalar, vector, and, in the case of XM, matrix processing, allowing devices to run AI models faster while keeping power and footprint low. Vector-based RISC-V CPUs process multiple data items in parallel, reducing instruction overhead and energy consumption compared to scalar-only CPUs, and the XM engine delivers scalable matrix performance. This family is aimed at companies building AI devices, particularly edge applications like cameras, robotics, and IoT devices, helping them integrate high-performance AI outside data centres.

SiFive

<https://www.sifive.com>

M.2 form factor

Edge AI applications, such as large language models (LLMs), vision transformers, and real-time computer vision, require high compute performance but face limitations on size, power, and thermal capacity. Cloud



processing also introduces latency, cost, and security risks. Axelera

AI's Metis M.2 Max addresses this by delivering PCIe-level AI performance in an M.2 form factor for laptops, edge devices, and industrial systems. Compared to the previous Metis M.2, it doubles memory bandwidth, reduces profile, and adds thermal management and security features. It provides a 33% performance boost for convolutional neural networks (CNNs) and doubles token throughput for LLMs and vision-language models (VLMs). Its 27% lower height allows deployment in tight spaces, and an onboard power probe automatically adjusts performance for power or thermal limits, making it suitable for industries such as industrial manufacturing, retail, security, healthcare, and public safety.

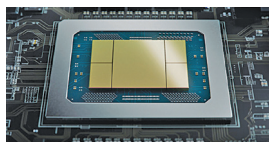
Axelera AI

<https://axelera.ai>

Bi-directional D2D interconnect

Chip designers and data centre operators face pressure to boost performance while reducing power use and die

size, as next-generation XPU's need higher bandwidth but conventional interconnects consume more energy and occupy more silicon area. Marvell Technology addresses this with the



first 2nm 64 Gbps bi-directional die-to-die (D2D) intercon-

nect, enabling 32 Gbps two-way connectivity over a single wire and reducing compute die area by 85% compared to traditional designs. It also features adaptive power management that matches device activity to bursty data centre traffic, cutting power use by up to 75% under normal workloads and up to 42% during peak traffic.

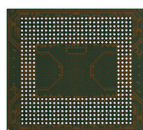
The interconnect provides bandwidth, energy efficiency, and smaller die footprints, helping designers and operators build faster, more efficient, and scalable processors for next-generation data centres.

Marvell

<https://www.marvell.com>

Mobile DRAM

SK hynix has introduced mobile DRAM with improved heat dissipation using High-K Epoxy moulding compound, addressing performance issues in smart-



phones caused by heat from on-device AI data

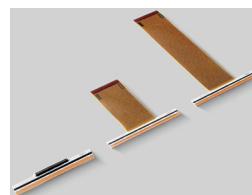
transfers. Most flagship smartphones use a package on package (PoP) structure that stacks DRAM on top of the mobile application processor (AP) to save space and increase data transfer speed, but this design traps heat from the AP inside the DRAM, reducing performance. The new DRAM increases thermal conductivity 3.5 times and reduces vertical thermal resistance by 47%, which improves device performance, lowers power consumption, extends battery life, and increases product lifespan. Smartphone makers are expected to benefit, as this technology helps prevent overheating in devices while supporting AI applications.

SK Hynix

<https://www.skhynix.com>

Printheads

Retail and service businesses, such as convenience stores and restaurants, require fast and reliable printing across various media, and the TPA series printheads from Kyocera meet these needs. They print at 350mm/sec while maintaining consistent quality and support expanded head mounting



tolerance for flexible integration. Improvements to the printhead structure and heat conduc-

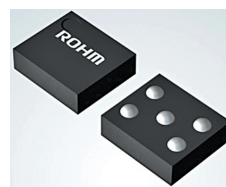
tion ensure stable printing across the full width on various media, while Kyocera's film formation technology increases corrosion and wear resistance for long life under high-volume conditions. The TPA series also supports Blue4est paper for phenol-free printing, providing performance with sustainability, making it a solution for operators who require fast, durable, and versatile printing in retail and service environments.

Kyocera

<https://global.kyocera.com>

CMOS op-amp

ROHM's TLR1901GXZ CMOS operational amplifier (op-amp) addresses the challenge of delivering sensing in



size- and power-constrained devices, such as handheld measurement instruments, wearable devices, and

indoor motion detectors. The op-amp features a low input offset voltage, approximately 45% lower than that of typical ultra-low current op-amps, enabling precise measurements. The combination of small size, low power, and accuracy makes the TLR1901GXZ suitable for consumer devices like wearables, smart devices, and motion sensors, as well as industrial equipment, including gas detectors, fire alarms, handheld measurement instruments, and environmental IoT sensors.

ROHM

<https://www.rohm.com>